

A root-split subcase of the height ω_2 Valdivia tree problem

Run `fa.banach.001`; source arXiv:2305.11737

Source problem

Russo and Somaglia's paper *Banach spaces of continuous functions without norming Markushevich bases* studies compact trees in the coarse wedge topology. After proving that scattered compact r -trees of height less than ω_2 are Valdivia, they ask the following boundary question.

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Problem 5.9. *Let T be a scattered, compact r -tree with $\text{ht}(T) = \omega_2$. Must T be Valdivia?*

In the light of the previous results, it is natural to wonder when compact trees are scattered. Therefore, we now proceed to give a characterisation of scattered trees. Given

Figure 1: Russo–Somaglia, arXiv:2305.11737, page 15: Problem 5.9.

The problem asks whether every scattered compact r -tree T with $\text{ht}(T) = \omega_2$ must be Valdivia.

Definitions

We use the source paper's terminology. A compact tree is always endowed with the coarse wedge topology. For $t \in T$, $V_t = \{u \in T : t \leq u\}$, and $\text{ims}(t)$ is the set of immediate successors of t . A tree is an r -tree if $|\text{ims}(t)| < \omega$ whenever $\text{cf}(t) \geq \omega_1$.

Proof intuition

The source paper already proves the desired Valdivia conclusion below the boundary height ω_2 . Therefore, if a height- ω_2 tree splits at the root into clopen pieces whose individual heights are all below ω_2 , each piece is already Valdivia. What remains is only to check that adding one compactifying root to a disjoint sum of Valdivia compacta preserves Valdivia compactness. This is done by putting the Valdivia embeddings on disjoint coordinate sets and using one extra coordinate to mark which summand a point belongs to.

The partial theorem

Theorem 1. *Let T be a rooted scattered compact r -tree with root 0_T and $\text{ht}(T) = \omega_2$. Suppose that*

$$\text{ht}(V_s) < \omega_2 \quad (s \in \text{ims}(0_T)).$$

Then T is Valdivia.

Proof. For every $s \in \text{ims}(0_T)$, the set V_s is clopen in T , with the subspace topology equal to the coarse wedge topology of the induced rooted tree. It is again a scattered compact r -tree. By hypothesis $\text{ht}(V_s) < \omega_2$, so Russo–Somaglia's Corollary 5.7 [1] implies that every V_s is Valdivia.

The tree T is the one-point compactification of the topological disjoint sum

$$\bigoplus_{s \in \text{ims}(0_T)} V_s,$$

where the compactifying point is 0_T . Indeed, the V_s 's are pairwise disjoint clopen subsets of T , their union is $T \setminus \{0_T\}$, and the neighborhoods of 0_T are exactly the sets containing

$$T \setminus \bigcup_{s \in F} V_s$$

for finite $F \subseteq \text{ims}(0_T)$.

It remains to use the following elementary stability lemma.

Lemma 1. *Let $(K_i)_{i \in I}$ be any family of Valdivia compact spaces. The one-point compactification of their topological disjoint sum is Valdivia.*

Proof. For each $i \in I$, choose a Valdivia embedding

$$h_i : K_i \longrightarrow [0, 1]^{\Gamma_i},$$

where the coordinate sets Γ_i are pairwise disjoint and $h_i(K_i) \cap \Sigma(\Gamma_i)$ is dense in $h_i(K_i)$. Add a new coordinate q_i for each i , and put

$$\Gamma = \bigcup_{i \in I} \Gamma_i \cup \{q_i : i \in I\}.$$

Let K be the one-point compactification and call its point at infinity ∞ . Define $H : K \rightarrow [0, 1]^\Gamma$ by $H(\infty) = 0$, and for $x \in K_i$,

$$H(x)(q_i) = 1, \quad H(x)|_{\Gamma_i} = h_i(x),$$

with all other coordinates equal to 0.

This map is injective and continuous on each summand. It is continuous at ∞ because a net that eventually leaves every finite collection of summands is eventually zero on each fixed coordinate. Conversely, the coordinate q_i separates the i -th summand from all other summands and from ∞ , so the induced topology is precisely the one-point compactification topology.

Finally,

$$\{\infty\} \cup \bigcup_{i \in I} \{x \in K_i : h_i(x) \in \Sigma(\Gamma_i)\}$$

is dense in K , and every one of its points has countable support in Γ . Hence $H(K) \cap \Sigma(\Gamma)$ is dense in $H(K)$, so K is Valdivia. $\square$

Applying the lemma to the family $(V_s)_{s \in \text{ims}(0_T)}$ proves that T is Valdivia. $\square$

Limitations

This is only a root-splitting subcase of Problem 5.9. It does not address the case in which some immediate successor subtree V_s still has height ω_2 , nor a height- ω_2 core that survives every finite clopen cut below the root.

The loop also inspected the source's separate Problem 5.16, asking whether $C(T)$ has a norming Markushevich basis for every tree T of height at most ω_1 . A direct edge-difference biorthogonal construction is tempting, but at countable-limit predecessor levels the resulting functional span is not visibly uniformly norming. No packet is claimed for Problem 5.16.

Verification and novelty notes

The proof uses only Russo–Somaglia Corollary 5.7 and the elementary stability lemma above. The source paper itself does not state this root-splitting subcase of Problem 5.9.

Before promotion, the lightweight run indexes were searched for 2305.11737, Valdivia, omega_2, compact r-tree, norming M-basis, and related tree-height phrases. No duplicate packet was found. Web searches on June 16, 2026 for the exact source title, arXiv id, and phrases around Problem 5.9 found the source paper but no later full answer or duplicate of this subcase.

Human review recommendation

Review the identification of T with the one-point compactification of the root successor subtrees, and the coordinate proof of the stability lemma. The remaining mathematical input is exactly Russo–Somaglia Corollary 5.7.

References

- [1] T. Russo and J. Somaglia, *Banach spaces of continuous functions without norming Markushevich bases*, arXiv:2305.11737.